

Exercise 8



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Macroeconomics I

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RBC: EXTENSIONS

Consider an economy similar to the one you see in project 1. Assume, however, that the household preferences takes the form

$$u(c_t, N_t) = \frac{(C_t v(1 - N_t))^{1-\sigma} - 1}{1 - \sigma}$$

where

$$v(1 - N_t) = \exp\left(\frac{\theta(1 - N_t)^{1-\zeta} - 1}{1 - \zeta}\right)$$

Assume:

$$\alpha = 1/3, \rho_A = 0.95, \delta = 0.025, \beta = 0.99, \theta = 0.5, \sigma = \zeta = 2. \quad (1)$$

Find the set of FOCs describing the competitive equilibrium of the system, and use Dynare to plot IRFs for Y, c, N, I, w, r . Compare these with your results for the case of quasi-linear utility representation function. Intuitively explain the differences between them.

(Household's Problem)

The representative household maximizes its lifetime utility function subject to it budget constraint:

$$\max_{c_t, k_{t+1}, N_t} U = \mathbb{E}_t \left[\sum_{i=0}^{\infty} \beta^i \left(\frac{(c_t v(1 - N_t))^{1-\sigma} - 1}{1 - \sigma} \right) \right], \quad (2)$$

$$\text{where: } v(1 - N_t) = \exp \left(\frac{\theta(1 - N_t)^{1-\zeta} - 1}{1 - \zeta} \right), \quad (3)$$

$$\text{s.t. } c_t + k_{t+1} \leq w_t N_t + (1 + R_t - \delta)k_t + \Pi_t. \quad (4)$$

The Lagrangian for the household's problem is:

$$\mathcal{L} = \mathbb{E}_t \sum_{i=0}^{\infty} \beta^i \left[\left(\frac{(c_t v(1 - N_t))^{1-\sigma} - 1}{1 - \sigma} \right) - \mu_t (c_t + k_{t+1} - w_t N_t - (1 + R_t - \delta)k_t - \Pi_t) \right]. \quad (5)$$

The FOCs are:

$$\frac{\partial \mathcal{L}}{\partial c_t} = 0 \implies c_t^{-\sigma} v(1 - N_t)^{1-\sigma} = \mu_t, \quad (6)$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \implies -c_t^{1-\sigma} v'(1 - N_t) v(1 - N_t)^{-\sigma} = \mu_t w_t, \quad (7)$$

$$\frac{\partial \mathcal{L}}{\partial k_{t+1}} = 0 \implies \mu_t = \beta \mathbb{E}_t [\mu_{t+1} (1 + R_{t+1} - \delta)], \quad (8)$$

$$\frac{\partial \mathcal{L}}{\partial \mu_t} = 0 \implies c_t + k_{t+1} = w_t N_t + (1 + R_t - \delta)k_t + \Pi_t. \quad (9)$$

We had:

$$v(1 - N_t) = \exp \left(\frac{\theta(1 - N_t)^{1-\zeta} - 1}{1 - \zeta} \right)$$

Which implies:

$$v'(1 - N_t) = -\theta(1 - N_t)^{-\zeta} \exp \left(\frac{\theta(1 - N_t)^{1-\zeta} - 1}{1 - \zeta} \right)$$

Substituting into FOCs, we get the following conditions:

$$\frac{\partial \mathcal{L}}{\partial c_t} = 0 \implies c_t^{-\sigma} \exp \left(\frac{\theta(1 - N_t)^{1-\zeta} - 1}{1 - \zeta} \right)^{1-\sigma} = \mu_t, \quad (\text{i})$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \implies c_t^{1-\sigma} \theta(1 - N_t)^{-\zeta} \exp \left(\frac{\theta(1 - N_t)^{1-\zeta} - 1}{1 - \zeta} \right)^{-\sigma+1} = \mu_t w_t, \quad (\text{ii})$$

$$\frac{\partial \mathcal{L}}{\partial k_{t+1}} = 0 \implies \mu_t = \beta \mathbb{E}_t [\mu_{t+1} (1 + R_{t+1} - \delta)], \quad (\text{iii})$$

$$\frac{\partial \mathcal{L}}{\partial \mu_t} = 0 \implies c_t + k_{t+1} = w_t N_t + (1 + R_t - \delta) k_t + \Pi_t. \quad (\text{iv})$$

From (i), (ii), we get intertemporal optimality condition:

$$c_t \theta(1 - N_t)^{-\zeta} = w_t, \quad (\text{v})$$

From (i), (iii), we get intratemporal optimality condition:

$$c_t^{-\sigma} \exp \left(\theta(1 - N_t)^{1-\zeta} - 1 \right)^{1-\sigma} = \beta \mathbb{E}_t \left[c_{t+1}^{-\sigma} \exp \left(\theta(1 - N_{t+1})^{1-\zeta} - 1 \right)^{1-\sigma} (1 + R_{t+1} - \delta) \right] \quad (\text{vi})$$

(Firm's Problem)

The firm's technology is described by the production function:

$$Y_t = z_t k_t^\alpha N_t^{1-\alpha} \quad (1)$$

The technology follows an *AR1* process:

$$\log(z_{t+1}) = \rho \log(z_t) + \epsilon_t, \quad \epsilon_t \sim IID \quad (2)$$

Firm wants to maximize its profit:

$$\max \quad \Pi_t = z_t k_t^\alpha N_t^{1-\alpha} - w_t N_t - R_t k_t \quad (3)$$

The optimality conditions are:

$$MP_L = (1 - \alpha) z_t k_t^\alpha N_t^{-\alpha} = w_t \quad (vii)$$

$$MP_K = \alpha z_t k_t^{\alpha-1} N_t^{1-\alpha} = R_t \quad (viii)$$

The structural equations of the economy are:

$$\text{(LNS, MARKET CLEARING)} \quad c_t + k_{t+1} = w_t N_t + (1 + R_t - \delta) k_t + \Pi_t \quad (ix)$$

$$\text{(PRODUCTION OF ECONOMY)} \quad Y_t = z_t k_t^\alpha N_t^{1-\alpha} \quad (x)$$

(The Steady State)

In the steady state we have no shock and zero profit ($\Pi = 0$, $z = 1$). Also the sytem parameters has no dynamic. So the equations are:

Supply:

$$c \theta (1 - N)^{-\zeta} (1 - \sigma) = w, \quad (\text{i})$$

$$c^{-\sigma} = \beta [(1 + R - \delta)]. \quad (\text{ii})$$

Demand:

$$(1 - \alpha) k^\alpha N^{-\alpha} = w, \quad (\text{iii})$$

$$\alpha k^{\alpha-1} N^{1-\alpha} = R. \quad (\text{iv})$$

Structural equations:

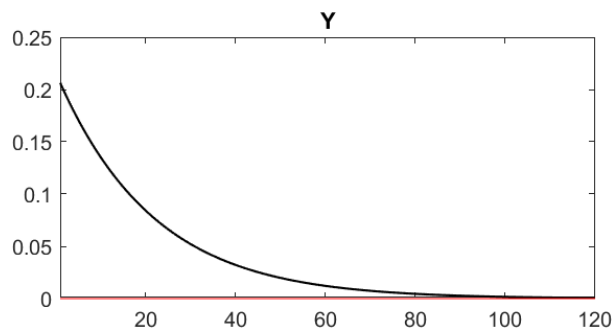
$$Y = C + I, \quad (\text{v})$$

$$Y = k^\alpha N^{1-\alpha} \quad (\text{vi})$$

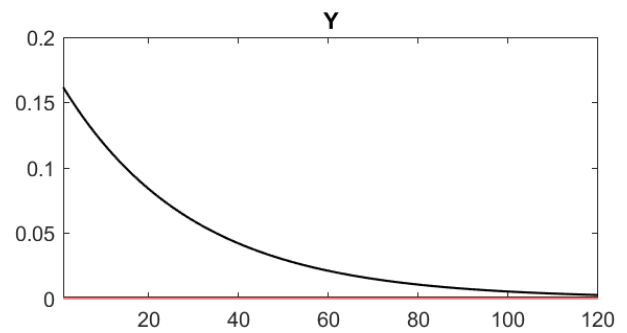
$$\log(z) = \rho \log(z) \quad (\text{vii})$$

$$k = k(1 - \delta) + i \quad (\text{viii})$$

(IRF Comparison)

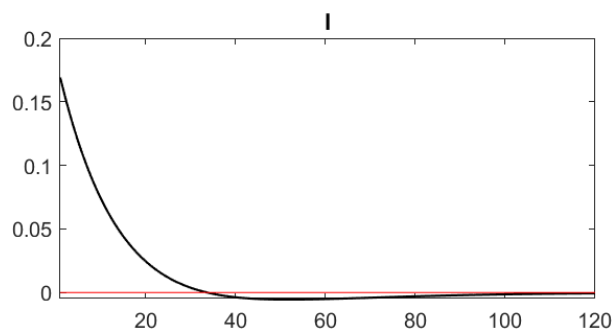


(a) Quasi-Linear Utility

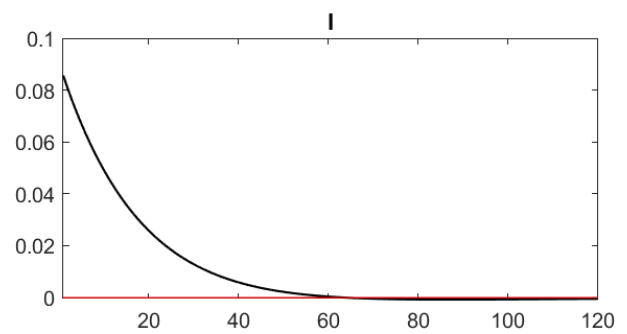


(b) Non-Seperable Utility

Figure 1: Output's IRF Comparison

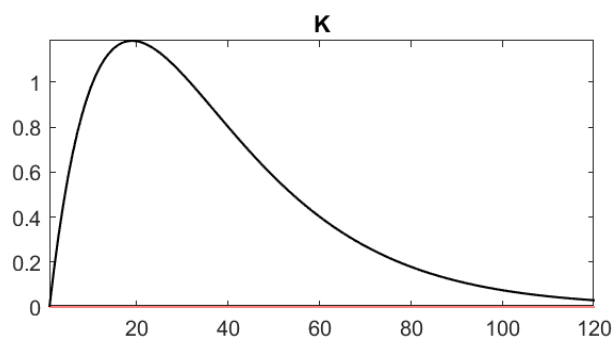


(a) Quasi-Linear Utility

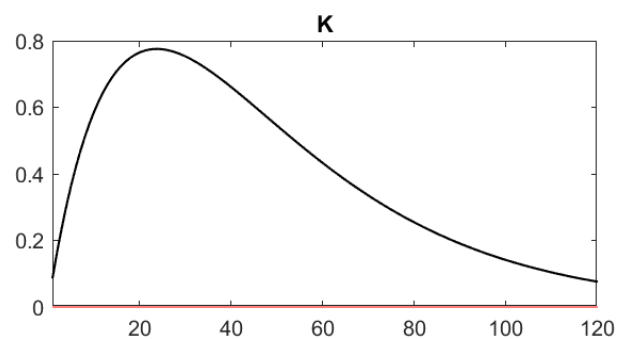


(b) Non-Seperable Utility

Figure 2: Investment's IRF Comparison

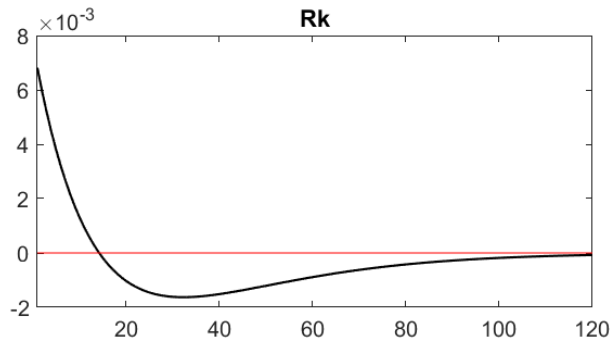


(a) Quasi-Linear Utility

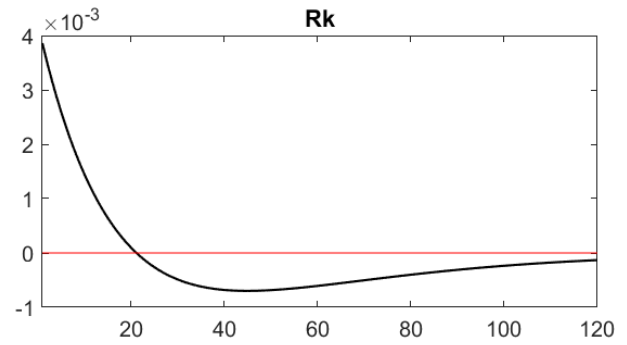


(b) Non-Seperable Utility

Figure 3: Capital's IRF Comparison

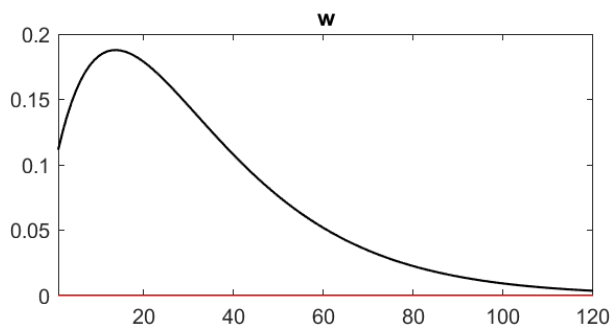


(a) Quasi-Linear Utility

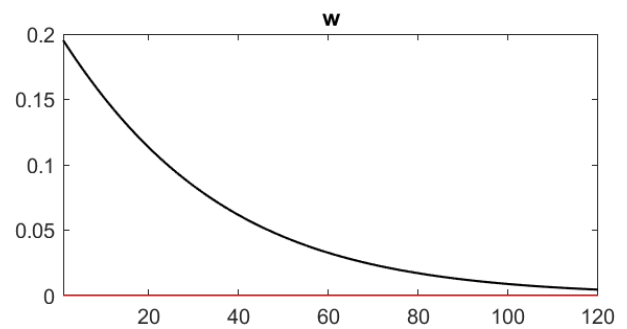


(b) Non-Seperable Utility

Figure 4: Interest Rate's IRF Comparison

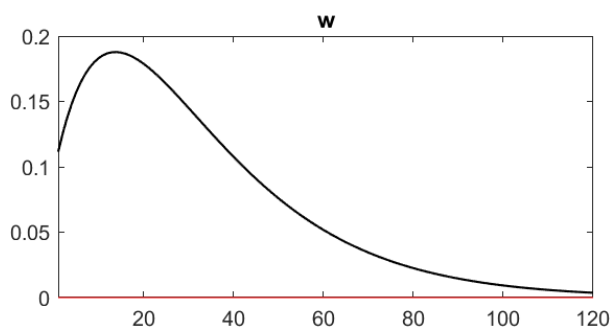


(a) Quasi-Linear Utility

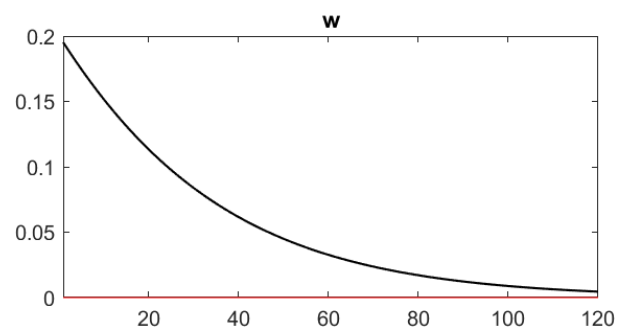


(b) Non-Seperable Utility

Figure 5: Technology's IRF Comparison



(a) Quasi-Linear Utility



(b) Non-Seperable Utility

Figure 6: Wage's IRF Comparison

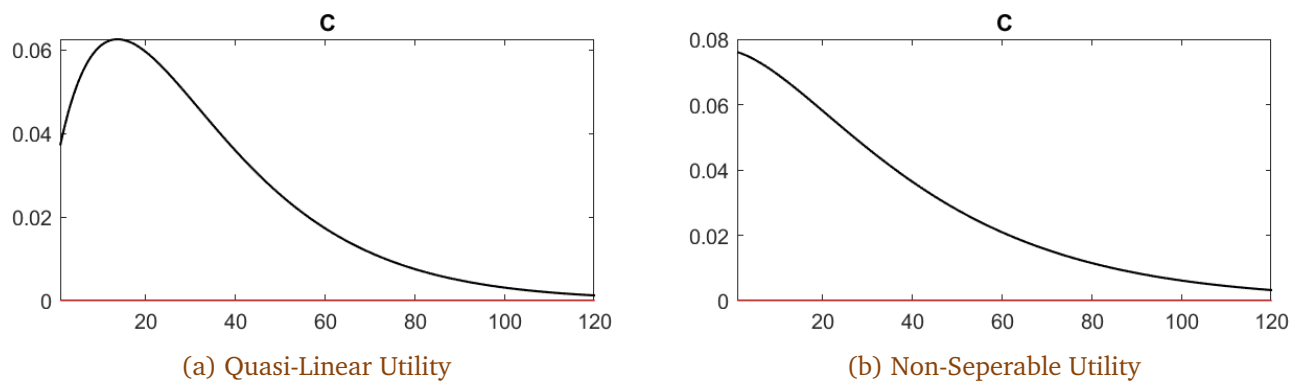


Figure 7: Consumption's IRF Comparison

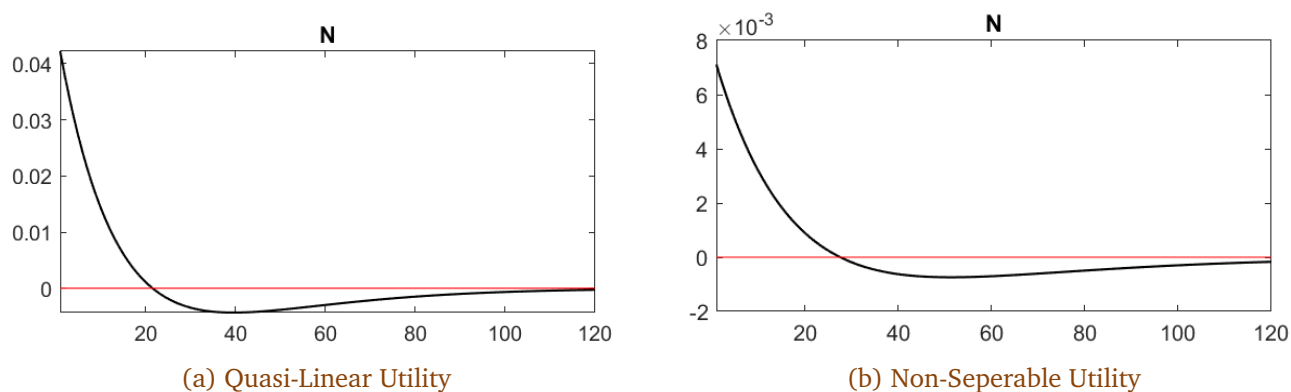


Figure 8: Labor's IRF Comparison

To open the Dynare script for this problem:

[Click here to open the code utilizing steady_state_model method.](#)

Or alternatively,

[Click here to open the code utilizing initval method.](#)

INTRODUCTION TO DYNAMIC PROGRAMMING

Your utility representation function is

$$u(c) = \frac{c^{1-\sigma}}{1-\sigma}. \quad (1)$$

Initially you are endowed with 1 unit of cake. Your problem is choosing the optimal consumption path to maximize the present value of the sum of your utility gains from eating cake. Formally, this would be

$$\max_{c_t, x_{t+1}} \sum_{t=0}^{\infty} \beta^t u(c_t), \quad (2)$$

subject to

$$x_{t+1} \leq x_t - c_t \quad \text{and} \quad 0 \leq c_t \leq x_t, \quad (3)$$

with $x_0 = 1$.

1. Bellman Equation

Write the Bellman equation of this problem.

The problem is to:

$$\max_{c_t, x_{t+1}} \sum_{t=0}^{\infty} \beta^t u(c_t), \quad (1)$$

subject to the constraint:

$$x_{t+1} \leq x_t - c_t \quad \text{and} \quad 0 \leq c_t \leq x_t, \quad (2)$$

with the initial condition $x_0 = 1$.

The problem is equivalent to finding the value function $v(x_t)$ that satisfies:

$$v(x_t) = \max_{c_t \in [0, x_t]} \{u(c_t) + \beta v(x_{t+1})\}, \quad (3)$$

where $u(c_t) = \frac{c_t^{1-\sigma}}{1-\sigma}$, and the constraint $c_t = x_t - x_{t+1}$ holds, with $x_0 = 1$.

Thus, the Bellman equation is:

$$v(x_t) = \max_{x_{t+1} \in [0, x_t]} \left\{ \frac{(x_t - x_{t+1})^{1-\sigma}}{1-\sigma} + \beta v(x_{t+1}) \right\}. \quad (4)$$

2. Value Function

Show that the value function of this problem is

$$v^*(x_t) = \left(1 - \beta^{\frac{1}{\sigma}}\right)^{-\sigma} u(x_t).$$

To show that the value function is

$$v^*(x_t) = \left(1 - \beta^{\frac{1}{\sigma}}\right)^{-\sigma} u(x_{t+1}), \quad (1)$$

we need to verify that this function satisfies the Bellman equation.

Substituting $v^*(x_t)$ into the Bellman equation, we get:

$$\max_{x_{t+1} \in [0, x]} \left\{ u(x_t - x_{t+1}) + \beta \left(1 - \beta^{\frac{1}{\sigma}}\right)^{-\sigma} u(x_{t+1}) \right\} \stackrel{?}{=} \left(1 - \beta^{\frac{1}{\sigma}}\right)^{-\sigma} u(x_{t+1}). \quad (2)$$

Taking the FOC from the Bellman equation yields:

$$\begin{aligned} \frac{d}{dx_{t+1}} v(x_t) &= \frac{d}{dx_{t+1}} \left(u(x_t - x_{t+1}) + \beta \left(1 - \beta^{\frac{1}{\sigma}}\right)^{-\sigma} u(x_{t+1}) \right) \\ &= -u'(x_t - x_{t+1}) + \beta \left(1 - \beta^{\frac{1}{\sigma}}\right)^{-\sigma} u'(x_{t+1}) = 0 \\ \implies \left(\frac{x_t - x_{t+1}}{x_{t+1}} \right)^{-\sigma} &= \beta \left(1 - \beta^{\frac{1}{\sigma}}\right)^{-\sigma} \\ \implies \frac{x_t}{\beta^{-\frac{1}{\sigma}} \left(1 - \beta^{\frac{1}{\sigma}}\right) + 1} &= x_{t+1} \end{aligned}$$

Since $x_{t+1} \in [0, x_t]$, we have interior solution.

Substituting $x_{t+1} = \frac{x_t}{\beta^{-\frac{1}{\sigma}} \left(1 - \beta^{\frac{1}{\sigma}}\right) + 1}$ into the LHS of Bellman equation, we will reach the proposed value function

3. Value Function Iteration

Use value function iteration to numerically solve for v^* . Compare your results. ($\sigma = 3$, $\beta = 0.98$)

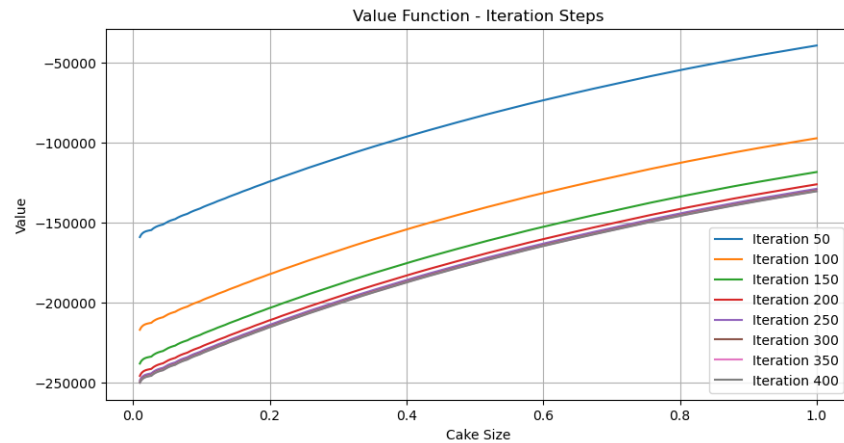


Figure 9: Iteration Plot

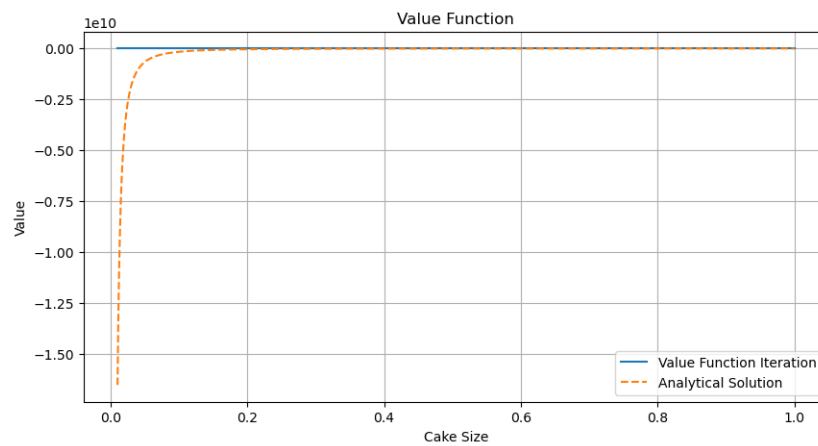


Figure 10: Analytical vs. Numerical Comparison

To open the .ipynb file for this problem [Click Here!](#)